

LiDAR based Road surface index Calculation

Project Team

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Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Hardware Setup	2
1.3.2	Data Collection	2
1.3.3	Data Processing	3
1.4	User Classes and Characteristics	4
2	Project Requirements	6
2.1	Functional Requirements	6
2.1.1	Hardware setup	6
2.1.2	Data Collection	6
2.1.3	Data Processing	7
2.2	Non-Functional Requirements	7
2.2.1	Reliability	7
2.2.2	Usability	7
2.2.3	Performance	8
2.2.4	Security	8
2.2.5	Scalability	8
2.2.6	Maintainability	8
2.3	Domain Model	8
2.4	Event Response Table	10
3	System Overview	11
3.1	Architectural Design	11
3.2	Design Models	11
3.3	Data Design	13
	References	15

List of Figures

2.1	Domain Model - RSI Calculation	9
3.1	Activity Diagram - RSI calculation	11
3.2	Activity Diagram - RSI calculation	12
3.3	ERD - RSI calculation	13
3.4	Class Diagram - RSI calculation	14

List of Tables

Chapter 1

Introduction

This document outlines the requirements for a road surface index calculation system that uses lidar sensors mounted on vehicles to assess the condition of road surfaces. The system is intended to be used by municipalities and road maintenance teams for monitoring and improving road quality. The document is designed for a variety of readers. Developers will focus on the technical aspects, such as system architecture, sensor integration, and software development. Project managers will be interested in the overall scope, timelines, and resource allocation. Testers will focus on validating the system's accuracy and reliability, while documentation writers will ensure the creation of comprehensive user guides and technical manuals.

1.1 Problem Statement

We are developing a road surface index calculation system to address the growing need for efficient, accurate, and automated road condition monitoring. According to regulations, the National Highway Authority (NHA) is required to conduct road surveys annually, but due to inefficiencies in current methods, they often survey roads only once every 3 to 5 years. This delay leads to worsening road conditions, resulting in higher repair costs and increased risks of accidents. Our software, using lidar sensors mounted on vehicles, offers a real-time, data-driven solution by automating the detection of road surface irregularities such as ruts, potholes and cracks. It calculates a comprehensive surface index, helping authorities prioritize repairs, optimize maintenance schedules, and allocate resources more effectively. By providing precise, continuous monitoring, the system improves road safety, extends infrastructure lifespan, and reduces overall costs, benefiting both authorities and road users.

1.2 Scope

The scope of our project is to develop a comprehensive road surface index calculation system that leverages lidar sensors and crowd sensing technology for efficient and accurate road condition monitoring. The primary functionality of the system includes detecting road surface irregularities such as ruts, cracks, and potholes, and calculating a detailed road surface index. This index will be used to prioritize repairs and schedule maintenance activities. The project will incorporate lidar sensors mounted on vehicles to gather real-time data on road surfaces, ensuring precise and continuous monitoring. In addition, we will integrate crowd sensing, allowing vehicles equipped with sensors to contribute in collections of road condition data, thereby increasing the system's coverage and enhancing data integrity. All collected data will be geotagged to ensure location accuracy, enabling authorities to easily identify and access specific problem areas on a map. The system will feature a user-friendly interface for road management authorities to visualize road conditions, generate reports, and manage maintenance priorities. The project will focus on developing algorithms for data processing, road condition classification, and integration of data collected through crowd-sensing to create a robust solution. The scope also includes testing and validating the system to ensure its reliability, scalability, and ease of deployment for large-scale road monitoring.

1.3 Modules

Below are the key modules we plan to develop in the first iteration of our project:

1.3.1 Hardware Setup

This involves setting up and calibrating the necessary hardware for data collection, ensuring the accuracy and efficiency of the system.

1. **Sensor Mounting:** Mount lidar sensors on vehicles for road data collection.
2. **Sensor Calibration:** Calibrate sensors to ensure accurate and consistent data collection across different terrains.

1.3.2 Data Collection

This module is responsible for gathering real-time road surface data using lidar sensors and crowd-sourced inputs. It ensures accurate and comprehensive coverage of road conditions.

1. **Lidar Integration:** Collect real-time road surface data using lidar sensors mounted on vehicles.
2. **Geotagging:** Automatically geotag all collected data to ensure precise location tracking of road conditions.
3. **Real-Time Data Transmission:** Transmit data to the server for immediate processing and analysis.

1.3.3 Data Processing

This module focuses on pre-processing the collected data to prepare it for analysis by machine learning models, which will be used to calculate the road surface index and detect road defects.

1. **Data Filtering:** Clean and filter the raw data to remove noise and irrelevant information. vehicles.
2. **Data Standardization:** Standardize data formats to ensure consistency for further analysis.
3. **Data Transmission:** Transmit the pre-processed data to the machine learning model for road surface index calculation and anomaly detection

1.4 User Classes and Characteristics

User class	Description
System Administrators	System Administrators are responsible for managing and maintaining the system, ensuring that it operates efficiently. They handle data backups, monitor system health, and ensure data privacy and security protocols are followed. Characteristics: High level of technical expertise, including experience in managing server infrastructure. Primary Tasks: Monitor and manage server resources. Perform system updates, patches, and backups. Ensure data is securely stored and accessible. Manage user accounts and permissions. Interaction with System: Access to the back-end server and data processing pipelines. Can configure the LiDAR sensor and ESP32 microcontroller remotely. Regularly access the database for monitoring incoming road defect data and performing audits.
Field Engineers	Field Engineers install, calibrate, and maintain the hardware components, such as LiDAR sensors and ESP32 microcontrollers, that are deployed on vehicles or road infrastructure. Characteristics: Strong technical skills in electronics, sensor technology, and hardware calibration. Familiarity with mobile networks for data transmission and hardware-to-software integration. Primary Tasks: Deploy LiDAR sensors on vehicles or road infrastructure. Troubleshoot and calibrate sensor devices. Ensure real-time data transmission from sensors to the server. Conduct on-site repairs and updates for hardware components. Interaction with System: Primarily interact with the hardware components (LiDAR and ESP32). Limited access to software components for calibration and real-time data testing. Ability to run diagnostics on hardware and connectivity issues.
Municipal Authorities	Municipal authorities, such as city officials or road maintenance supervisors, use the system to monitor road conditions and schedule repairs based on the road defect data. Characteristics: Non-technical users, generally familiar with basic digital tools. Require easy-to-understand reports and visual representations of road defects. Responsible for decision-making regarding road repairs and maintenance priorities. Primary Tasks: Review road defect data collected by the system. Prioritize road maintenance based on the severity of detected defects. Generate and analyze reports for resource allocation and repair scheduling. Interaction with System: Access the user interface to view road defect data in visual formats such as maps and charts. Receive notifications for critical road conditions. Ability to download and generate reports for meetings and decision-making.

User class	Description
Vehicle Owners (Crowd-Sensing Participants)	These users own vehicles equipped with LiDAR sensors for crowd-sensing purposes. They contribute to data collection by driving on city roads, allowing the system to gather defect information passively. Characteristics: Typically non-technical users with limited interaction with the system. Interested in the health of the roads they drive on but do not need detailed technical reports. May receive incentives (e.g., discounts, coupons) for participating in the crowd-sensing initiative. Primary Tasks: Drive their vehicles as part of their normal routine while the system collects road defect data passively. Receive updates or incentives for their participation in the system. Report sensor malfunctions or issues with the system. Interaction with System: Minimal interaction; the system operates automatically in the background. Can receive notifications if a sensor needs calibration or repair. May access a simplified mobile app or dashboard to check their contributions (e.g., miles covered, areas scanned).

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Functional Requirements

This section describes the functional requirements of the system.

2.1.1 Hardware setup

Following are the functional requirements for the hardware setup module:

1. **Sensor Mounting:** The system must mount lidar sensors securely on the vehicle to ensure stable data collection.
2. **Sensor Calibration:** The system must calibrate the lidar sensors before data collection to account for terrain variations.
3. **Weatherproofing:** The system must ensure that all hardware components are weatherproof and capable of functioning in varying environmental conditions.

2.1.2 Data Collection

Following are the functional requirements for the data collection module:

1. **Real-Time Data Capture:** The system must collect real-time road surface data using lidar sensors.
2. **Geotagging:** The system must geotag all collected data to ensure precise location tracking of road conditions.

3. **Data Transmission:** The system must transmit data to the server in real time for immediate processing.

2.1.3 Data Processing

Following are the functional requirements for the data processing module:

1. **Data Filtering:** The system must filter the collected data to remove noise and irrelevant information before analysis.
2. **Data Standardization:** The system must standardize data formats to ensure compatibility with the machine learning model.
3. **Model Input Preparation:** The system must transmit the pre-processed data to the machine learning model for further analysis and road surface index calculation.

2.2 Non-Functional Requirements

Following are the non-functional requirements for the system:

2.2.1 Reliability

The system must maintain a high level of reliability, with a target Mean Time Between Failures (MTBF) of 500 hours. A failure is defined as any instance where the system is unable to process or collect road surface data as intended. To protect against failures, the system will incorporate redundancy measures, such as backup sensors and data storage. An error detection strategy will involve real-time monitoring of system performance, and the correction strategy will include automated recovery procedures to restore functionality within 10 minutes of a failure.

2.2.2 Usability

The system shall be designed for ease of use, allowing users to complete primary tasks, such as data collection and analysis, with minimal training. For example, users should be able to retrieve previous road condition reports with a single interaction. Example:

USE-1: the interface must provide clear instructions and feedback during user interactions to enhance the overall user experience and minimize errors.

2.2.3 Performance

The system must prioritize accuracy in data processing and analysis over speed. The machine learning model should be designed to achieve a minimum accuracy rate of 90% in road surface index calculations and anomaly detection. While the time taken for processing will be monitored, it is acceptable for the model to require up to 30 seconds for thorough analysis, ensuring that the results are reliable and precise. The system will provide periodic updates on performance metrics to ensure that accuracy standards are consistently met.

2.2.4 Security

The system must ensure the protection of collected data against unauthorized access and breaches. Security requirements include limiting access to authorized personnel only, ensuring data encryption during transmission, and maintaining an audit trail for all data access and modifications. The system must also be resilient against common cybersecurity threats, such as SQL injection and denial-of-service attacks.

2.2.5 Scalability

The system must be designed to handle increasing amounts of data without a decline in performance. It should support the addition of more lidar sensors and crowd-sensing devices seamlessly, accommodating up to 100 vehicles collecting data simultaneously.

2.2.6 Maintainability

The system must be designed for easy maintenance and updates, with a goal of reducing downtime for software updates to less than 1 hour per month. The architecture should support modular upgrades to specific components without affecting overall system functionality.

2.3 Domain Model

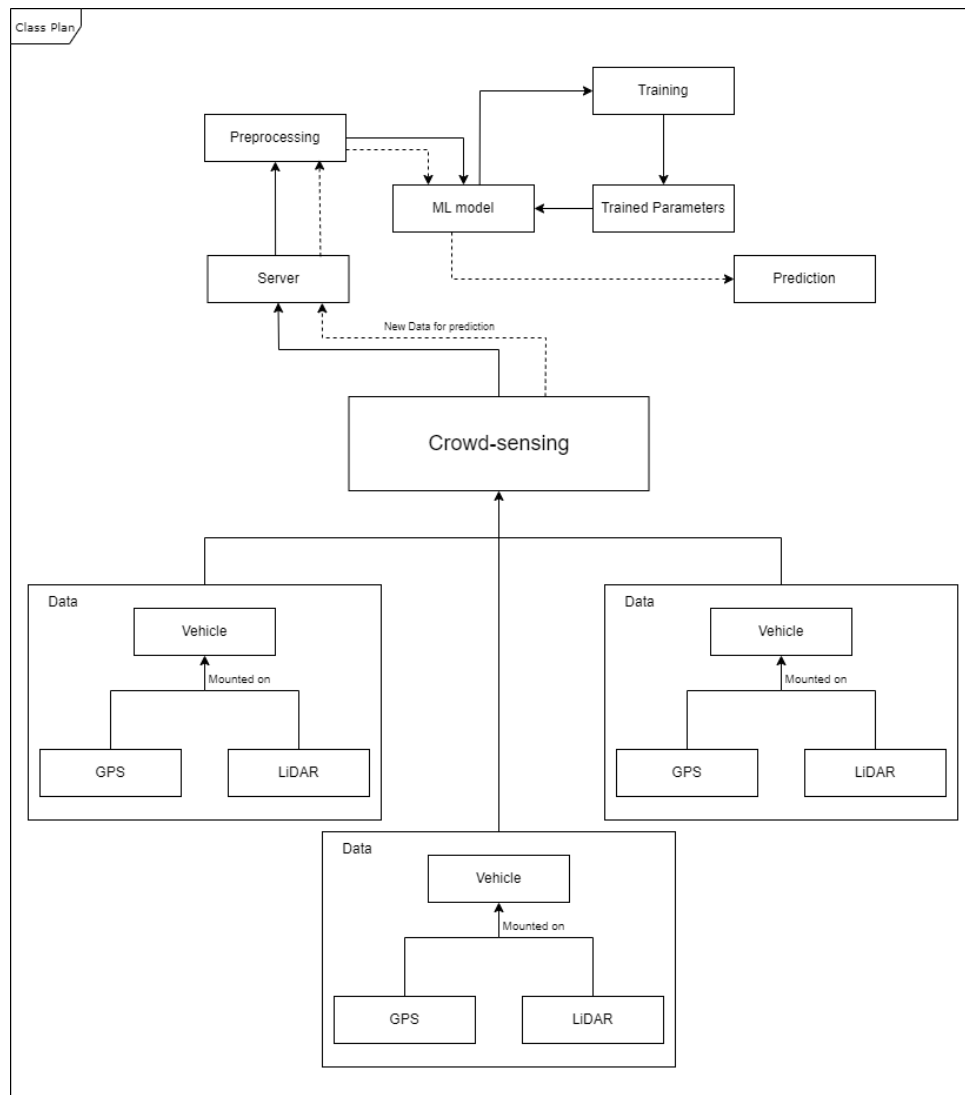


Figure 2.1: Domain Model - RSI Calculation

2.4 Event Response Table

Event	System Response
Lidar sensor transmits real-time road data	- Receive and preprocess the lidar data. - Update road condition database and calculate surface condition score.
End client opens the app and requests map view	- Fetch the latest heatmap data from the server. - Display the road surface condition scores for the user's region or specified area on the map.
Admin assigns a survey task	- Notify the relevant survey team of the new task with all details (location, priority, etc.). - Create an entry in the task management system for tracking.
Survey team updates task status	- Update the task's status in the database (e.g., "In Progress", "Completed"). - Notify the admin of any changes.
Survey team uploads task results	- Store the uploaded data (photos, notes, measurements) in the task database. - Update the task as completed and notify the admin.
Lidar sensor detects critical road condition	- Automatically generate an alert in the system. - Notify the admin for potential repair task assignment.
Admin assigns a repair task	- Notify the repair team with task details (location, priority, required materials, etc.). - Track the progress of the task in real-time.
Repair team updates task status	- Log the repair progress in the system. - Mark the task as completed once done and notify the admin.
User provides feedback on road conditions	- Receive and log feedback in the database. - Notify the admin for review and potential action.
System performs scheduled data aggregation	- Aggregate and analyze data from all connected lidar devices. - Update the road condition heatmap and calculate trends based on historical data.
System detects connection loss from a device	- Notify the end client of the connection issue. - Log the event and attempt to reconnect or troubleshoot automatically.

Chapter 3

System Overview

3.1 Architectural Design

Architecture

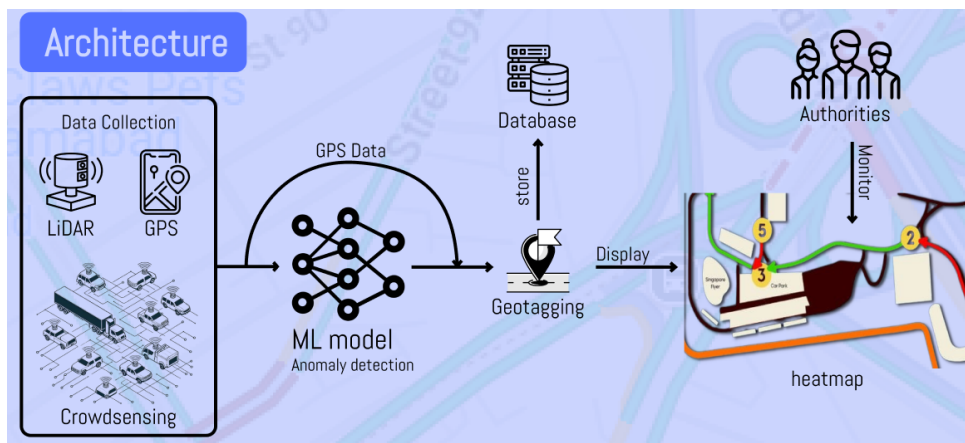


Figure 3.1: Activity Diagram - RSI calculation

3.2 Design Models

The applicable models for the project includes:

Activity Diagram

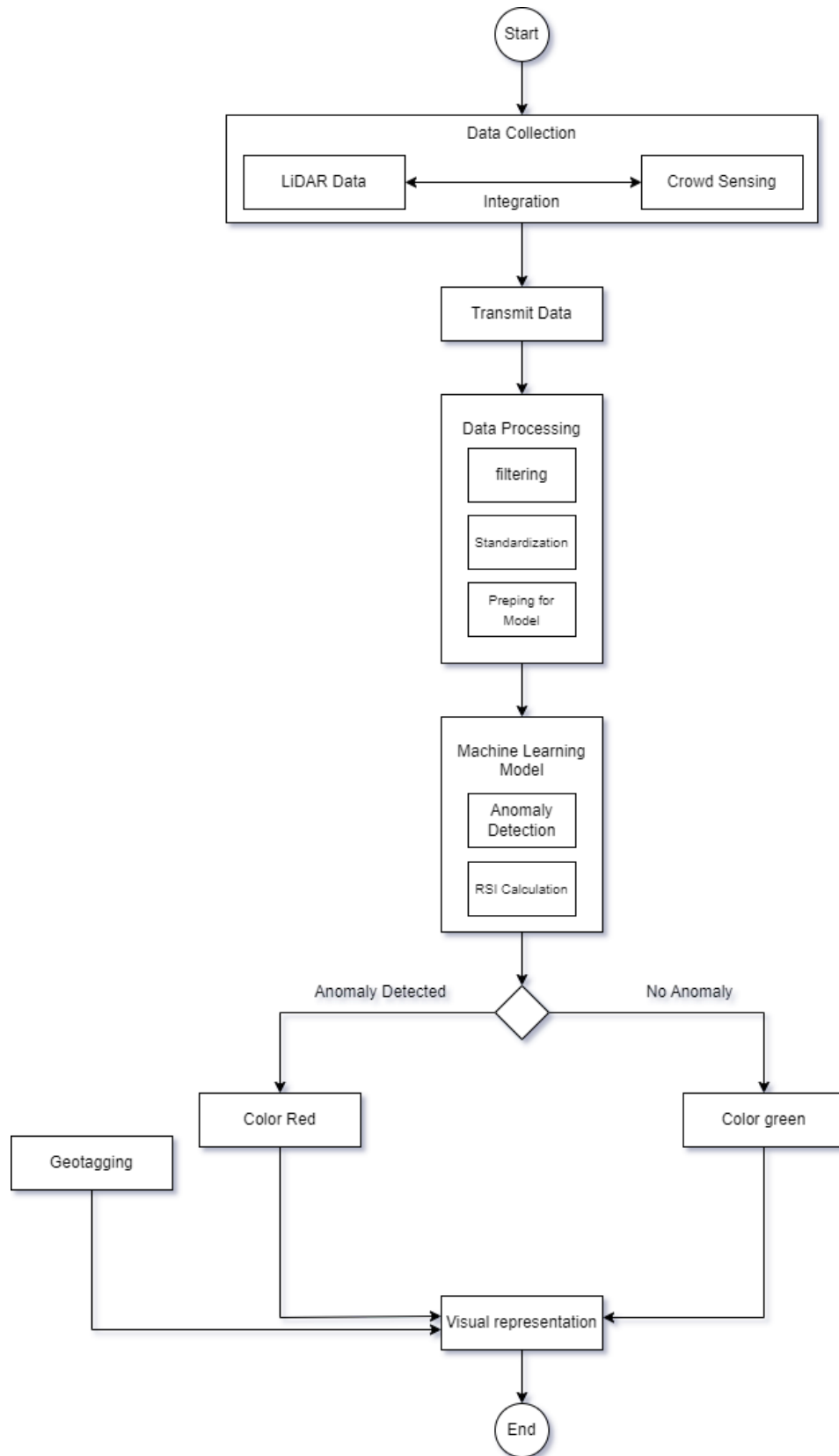


Figure 3.2: Activity Diagram - RSI calculation

3.3 Data Design

ERD

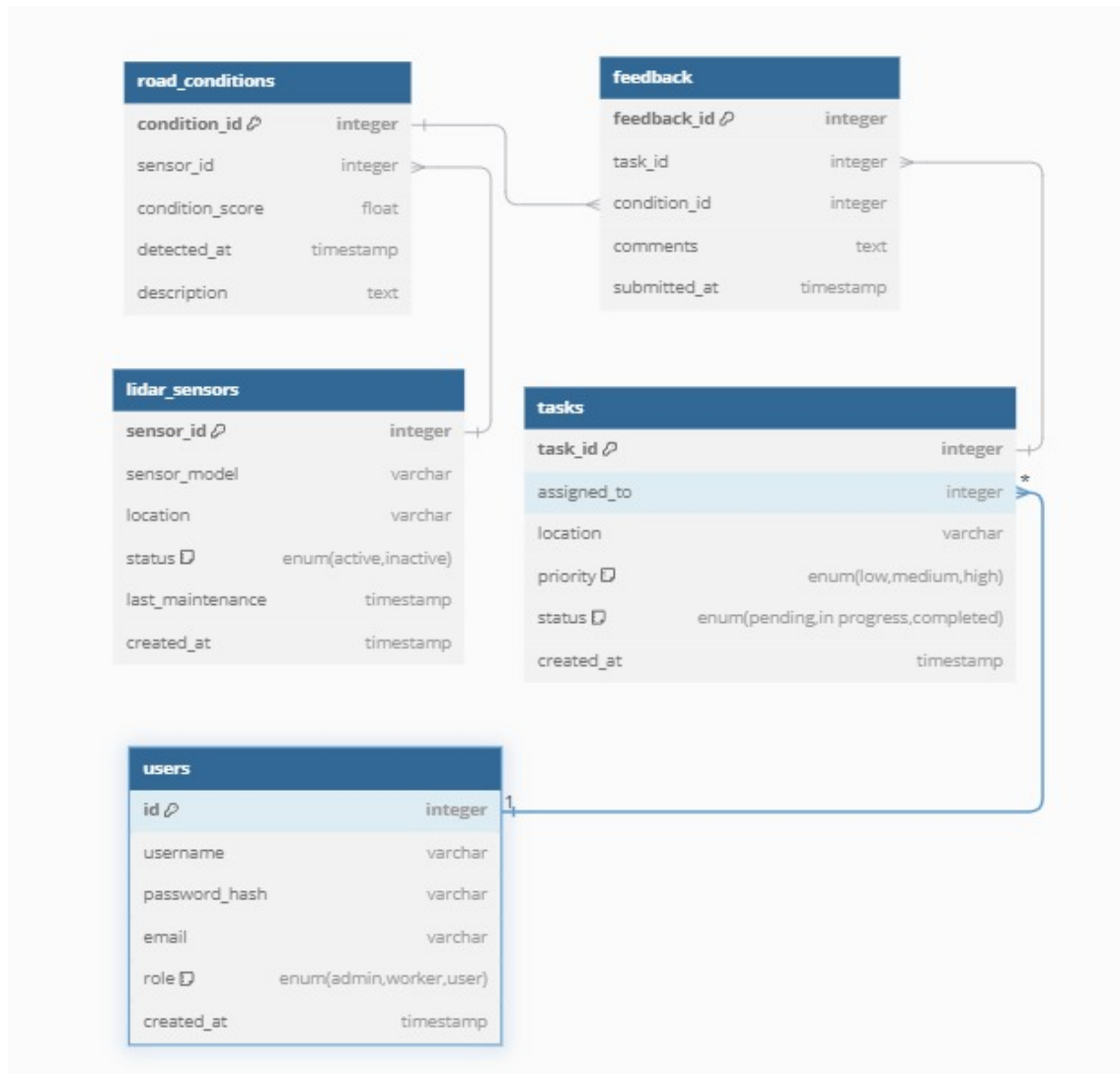


Figure 3.3: ERD - RSI calculation

Class Diagram

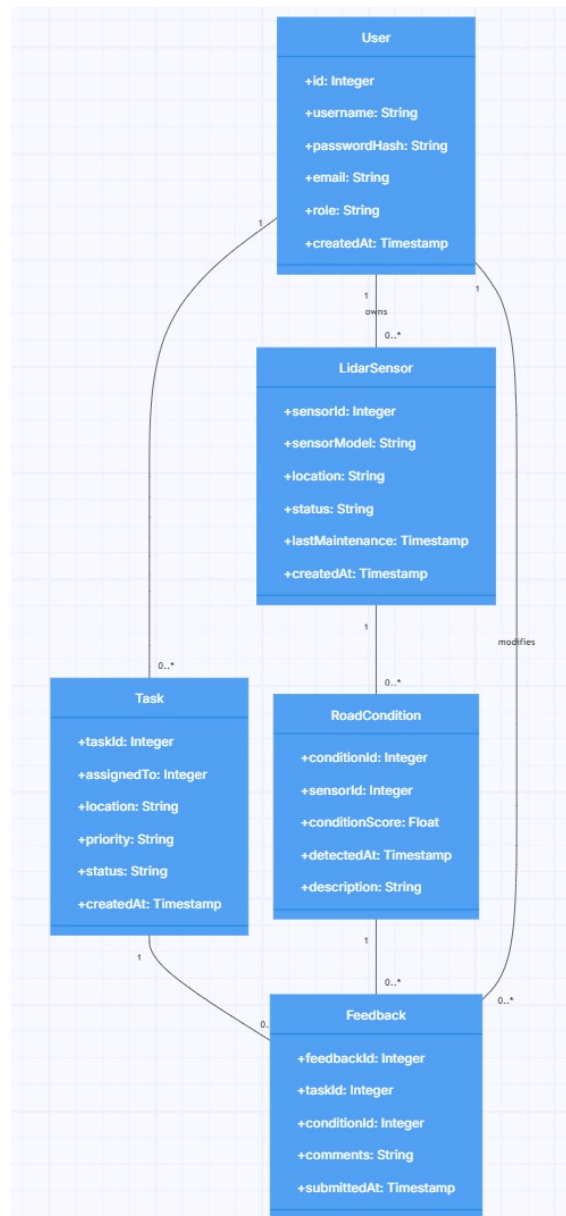


Figure 3.4: Class Diagram - RSI calculation

Bibliography